

Spectral Moments with Attractor Mass: A Similarity Index for One-Dimensional Cellular Automata

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Abstract. Classifications of cellular automata group rules by their behavior, but they do not measure how similar the rules are. We look for a similarity index based on the evolution that allows rules to be compared, even when they have different numbers of states or neighbors. First, we define four exact relations (state permutation, mirror, state reduction and neighborhood reduction), which reduce the 256 rules of the elementary cellular automata (ECA) to 81 classes. Then we compare the transition graph on a ring of n cells: its canonical form, together with the quotient by translations, distinguishes the 88 classes under permutation and mirror with only 6 cells, but the comparison is binary. As a gradual measure, we propose spectral moments with attractor mass, a fixed-length vector that, unlike the eigenvalues and the spectral moments, counts how many configurations reach each attractor and in how many steps. With the cosine distance, this vector distinguishes 86 of the 88 classes (87 together with the quotient), including rules 0 and 8, and gives zero distance to the relations that preserve the graph and, in the quotient, to those that differ by a translation, but it does not identify state reduction. Moreover, it depends on the size of the space, and it groups the ECA into four families according to the dominant period of their attractors.

Keywords: Cellular automata · Elementary cellular automata · Similarity index · Transition graph · Spectral moments · Basins of attraction.

1 Introduction

Cellular automata are discrete dynamical systems that have been used to model complex systems [1–3]: from very simple local rules they produce behaviors that range from convergence to a homogeneous, easily predictable state to chaotic dynamics [4, 5]. To distinguish these behaviors, several classifications have been proposed, such as Wolfram’s four classes [5], the classification of Li and Packard [6] and others reviewed by Martínez [7], or the set of *life-like* rules [8]. But how similar are the rules within the same class? And how different are they from the rules of another class?

Apart from directly comparing their exact behavior, classifications are the only measure to decide whether two rules are alike, and both give a binary answer. The purpose of this research is to define a method to compare the behavior of any two rules, even if they have different numbers of states or neighbors, and to measure their degree of similarity.

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We first identify the exact relations that exist between rules of different domains (Section 3) and use them as a reference to validate the proposed methods. We then compare the transition graph through its canonical form (Section 4) and propose spectral methods that represent each rule by a vector (Section 5): the eigenvalues, the spectral moments and the spectral moments with attractor mass. Finally, we evaluate this last method on rules from different domains, on spaces of different sizes, and by clustering the elementary rules (Section 6).

2 Preliminaries

A cellular automaton is a collection of cells arranged in a lattice, where each cell takes a state from a finite set. The system evolves in discrete time steps and, at each step, all cells update their state synchronously as a function of the states of their neighbors [4, 9, 10].

Definition 1. A cellular automaton is a 4-tuple $(\mathcal{L}, \mathcal{S}, \mathcal{N}, \mathcal{R})$ where:

- $\mathcal{L} \subseteq \mathbb{Z}^D$ is the D -dimensional cellular space; each cell is identified by its position $\vec{i} \in \mathcal{L}$.
- $\mathcal{S} = \{0, 1, \dots, q-1\}$ is the finite set of q states. The state of cell $\vec{i}$ at time t is denoted by $\sigma_{\vec{i}}(t) \in \mathcal{S}$.
- $\mathcal{N} = (\vec{v}_1, \dots, \vec{v}_m)$ is the ordered tuple of the m displacement vectors that define the neighborhood: the neighbors of $\vec{i}$ are the cells $\vec{i} + \vec{v}_j$, with $j = 1, \dots, m$.
- $\mathcal{R} : \mathcal{S}^m \rightarrow \mathcal{S}$ is the local rule, which assigns the new state of a cell from the states of its m neighbors:

$$\sigma_{\vec{i}}(t+1) = \mathcal{R} \left(\sigma_{\vec{i}+\vec{v}_1}(t), \dots, \sigma_{\vec{i}+\vec{v}_m}(t) \right).$$

The order of the tuple $\mathcal{N}$ matters: it determines the numbering of the rules and allows us to define relations such as mirror rules (Section 3.1).

2.1 Space, neighborhoods and rules

In practice, the cellular space is restricted by open boundaries, which assign a fixed state to the missing neighbors, or periodic ones, which take them from the opposite boundary [11, 12]. Throughout this work we always use periodic boundaries: in one dimension, the space is a ring of n cells, $\mathcal{L} = \mathbb{Z}_n$, and with radius r the neighborhood contains the $m = 2r + 1$ cells at distance at most r , including the cell itself.

The elementary cellular automaton (ECA) has $q = 2$, radius $r = 1$ ($m = 3$) and rule $\mathcal{R} : \{0, 1\}^3 \rightarrow \{0, 1\}$. There are $q^m = 2^3 = 8$ possible neighborhood patterns and each one is assigned a new state, so the total number of rules is $q^{q^m} = 2^8 = 256$. Each rule is identified by a decimal number obtained by reading, as a binary number, the states assigned to the patterns 111, 110, $\dots$, 000 [5]. In general, we denote by $r_{q,m}$ the rule number r with q states and m neighbors, numbered in the same way in base q ; thus, the ECA are the rules $r_{2,3}$. When m is even the neighborhood cannot be symmetric: with $m = 2$ we use the cells $(i-1, i)$, which is the *Mathematica* convention for radius $r = 1/2$.

2.2 Global function and transition graph

Let $C = \mathcal{S}^{\mathcal{L}}$ be the set of configurations, with $N = |C| = q^{|\mathcal{L}|}$. The global function $F : C \rightarrow C$ applies the local rule to all cells synchronously. The transition graph $G_F = (C, E)$ has one node per configuration and a directed edge $\sigma \rightarrow F(\sigma)$. Since F is a function, every node has out-degree 1 (a functional graph). Each connected component contains exactly one cycle, the attractor, with transient trees hanging from it; together they form its basin of attraction. For each $x \in C$ we define its period $p(x)$ as the length of the cycle it reaches, and its height $h(x)$ as the minimum number of steps needed to reach it ($h(x) = 0$ if x is periodic). Configurations without a predecessor are known as Garden of Eden configurations.

2.3 Quotient by translations

Let T be the translation by one position in $\mathcal{L}$ (a rotation of the ring $\mathbb{Z}_n$). With periodic boundaries, $F \circ T = T \circ F$, so two configurations that differ by a translation have equivalent trajectories. This allows us to build the quotient graph, whose nodes are the classes of configurations under translation and whose edges are $[x] \rightarrow [F(x)]$. All configurations in the same class have the same height, and the number of nodes drops to approximately $N/|\mathcal{L}|$. In the quotient graph each class counts as a single node, so the proportions computed on it are normalized by the number of classes.

3 Exact relations: Inheritance properties

This method extends the well-known equivalence criteria, state permutation and mirror rules, to rules of different spaces. For example, rule 0 always has the same behavior regardless of the number of states, the size of the neighborhood or the dimension. To formalize this, we call the domain of a rule the 3-tuple of dimension, states and neighborhood on which it is defined; two rules belong to the same domain if they share all three elements.

3.1 Rules in the same domain

In ECA, state permutation and mirror rules reduce the 256 rules to 88 classes [7]. Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be two cellular automata in the same domain and $(x_1, \dots, x_m) \in \mathcal{S}^m$ the state of the neighborhood. We say that they are equivalent if one of the following conditions holds:

- **State permutation:** there is a bijection $\pi : \mathcal{S} \rightarrow \mathcal{S}$ such that

$$\mathcal{R}_2(x_1, \dots, x_m) = \pi \left(\mathcal{R}_1 \left(\pi^{-1}(x_1), \dots, \pi^{-1}(x_m) \right) \right), \quad \forall (x_1, \dots, x_m) \in \mathcal{S}^m,$$

that is, if relabeling the states leaves the structure unchanged. For $\mathcal{S} = \{0, 1\}$, the only non-trivial permutation is $\pi(x) = 1 - x$.

- **Mirror rules:** let $M(x_1, \dots, x_m) = (x_m, \dots, x_1)$ be the reflection of the neighborhood; $\mathcal{A}_2$ is the mirror rule of $\mathcal{A}_1$ if $\mathcal{R}_2 = \mathcal{R}_1 \circ M$. That is, if we reflect the initial configuration, evolve it with $\mathcal{A}_1$ and reflect the result, we obtain the evolution of $\mathcal{A}_2$.

3.2 State reduction

Let $\mathcal{A}$ be a cellular automaton and $a \in \mathcal{S}$ one of its states. State reduction is possible when:

1. State a is never produced by the rule: $\mathcal{R}(x_1, \dots, x_m) \neq a, \forall (x_1, \dots, x_m) \in \mathcal{S}^m$.
2. Every neighborhood that contains state a always produces the same state b : $\mathcal{R}(x_1, \dots, x_m) = b$ if $x_j = a$ for some j .

Then the reduced automaton is $\mathcal{A}' = (\mathcal{L}, \mathcal{S}', \mathcal{N}, \mathcal{R}')$ with $\mathcal{S}' = \mathcal{S} \setminus \{a\}$ and $\mathcal{R}'$ the restriction of $\mathcal{R}$ to $\mathcal{S}'^m$. Since state a is never produced, it can only appear in the initial condition, and since every neighborhood that contains it produces b , it disappears in the first step; from then on, the evolution of $\mathcal{A}$ is exactly that of $\mathcal{A}'$. Conversely, every rule with q states can be extended to $q + 1$ states by adding a new state that is never produced and that sends every neighborhood in which it appears to 0.

3.3 Neighborhood reduction

Neighborhood reduction is possible when the state of a neighbor does not affect the transition, that is, there is an index j such that

$$\mathcal{R}(x_1, \dots, x_j, \dots, x_m) = \mathcal{R}(x_1, \dots, x'_j, \dots, x_m), \quad \forall (x_1, \dots, x_m) \in \mathcal{S}^m, \forall x'_j \in \mathcal{S}.$$

Then the reduced automaton is $\mathcal{A}' = (\mathcal{L}, \mathcal{S}, \mathcal{N}', \mathcal{R}')$ with $\mathcal{N}' = \mathcal{N} \setminus \{\vec{v}_j\}$. If the removed neighbor is not at one of the ends, the neighborhood is adjusted so that it is contiguous with $m - 1$ cells, under the hypothesis that the distance between the neighbors only transforms the space of the evolution and not the behavior.

Applying these four relations iteratively builds a relation graph between rules of different domains, whose connected components are the equivalence classes; in ECA, 81 classes remain. Although this classification is still binary, it serves as a reference to validate the following methods.

4 Structural comparison of transition graphs

The previous relations depend on the rule vector. As an alternative, we compare the behavior: on a periodic space of n cells, with $N = |\mathcal{S}|^n$ configurations, we evolve each configuration to obtain the transition graph G_F of the rule. As a parallel method, we consider all the translations of a configuration as a single node (Section 2.3), which yields a quotient graph with approximately $|\mathcal{S}|^n/n$ nodes. We then say that two rules have the same behavior if their transition graphs are isomorphic.

The isomorphism comparison confirms that the ECA rules related by state permutation or mirror have isomorphic graphs, which corroborates that the transition graph represents the similarity between rules. Since this process is computationally expensive, we use the canonical form of the graph: a topological encoding that does not depend on the node labels, so that two graphs are isomorphic if and only if their canonical forms are equal.

These methods have some restrictions: the comparison is binary, it is between rules of the same domain, and the result depends on the size n of the space (Figure 1). In ECA, the full graph distinguishes 82 classes with 4 cells and a maximum of 85 with 6. From $n = 8$ on, some patterns appear: even n groups $\{15,170\}$, $\{23,178\}$ and $\{77,232\}$; n multiple of 4 also groups $\{105,150\}$; odd n groups $\{10,34\}$, $\{24,46\}$ and $\{136,160\}$; $n = 7$ and $n = 23$ group $\{60,90\}$. Combining an even and an odd size greater than 5 yields the 88 classes. The quotient distinguishes 83 classes with 6 cells and a maximum of 85 with 10; from $n = 7$ on, the number depends only on $n \bmod 4$ (80 classes if n is odd, 85 if $n \equiv 2 \pmod{4}$ and 84 if $4 \mid n$), and for every $n \geq 6$ it groups $\{12,34\}$, $\{15,51\}$ and $\{170,204\}$.

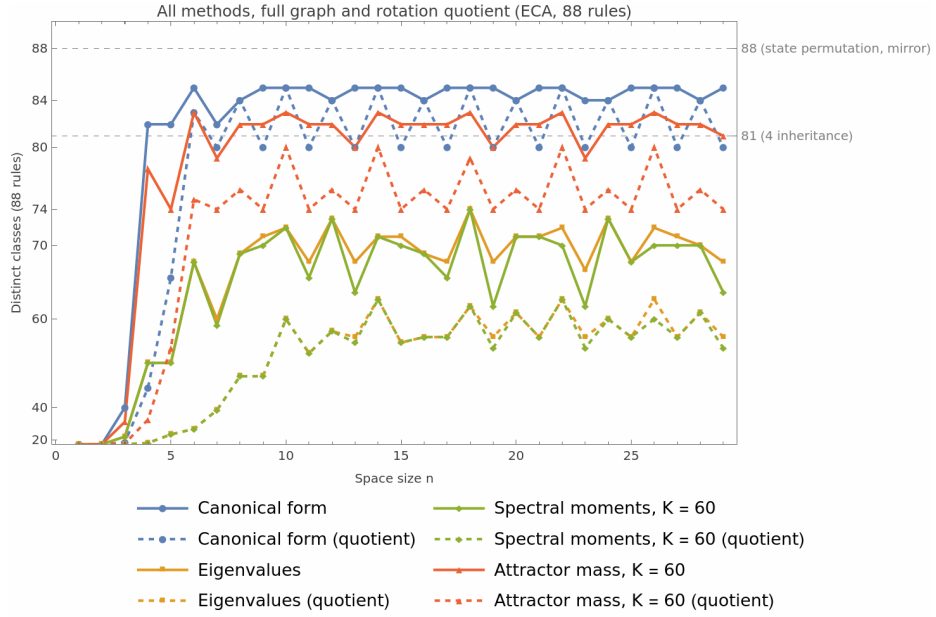


Fig. 1: Number of classes among the 88 representative rules with each method for $n = 1, \dots, 29$.

Using both canonical forms on the same space distinguishes the 88 behaviors with $n = 6, 10, 12, 14, 18, 22, 24$ and 26 . That is, a single space of 6 cells is enough if both the full graph and its quotient are used.

5 Spectral methods

To obtain a gradual measure, we represent each rule as a numerical vector (*embedding*) derived from its transition graph, so that the distance between vectors works as a similarity index.

5.1 Eigenvalues

We compute the eigenvalues of the adjacency matrix of the transition graph, sort them and use them as the *embedding* of the rule; two rules are compared through the cosine distance between their vectors, which measures how different their spectra are. This method has two limitations:

- **Blindness to transients:** each connected component is made of trees that lead to a cycle, and since the trees only contribute zero eigenvalues, the spectrum depends only on the attractors. ECA rules 0 and 8 have a single attractor, reached in at most 1 and 2 steps, respectively, so their eigenvalues are identical.
- **Domain dependence:** the number of eigenvalues equals the number of nodes $N = |S|^n$, so only rules with the same number of states and the same space size can be compared.

In the full graph, from $n = 8$ on we obtain between 67 and 74 classes (Figure 1). In the quotient, from $n = 12$ on we obtain between 56 and 63, and combining it with the full graph adds no classes (77 in both cases, Table 1). The groups $\{0,8\}$, $\{1,19\}$, $\{2,24,46\}$, $\{4,12\}$, $\{18,126\}$, $\{36,72\}$, $\{128,136\}$, $\{130,152\}$, $\{132,140\}$ and $\{162,168\}$ have the same spectrum for every size computed: they have the same cycles and differ only in their transients.

5.2 Spectral moments

To remove the domain dependence we propose to use spectral moments: instead of the full spectrum, for each step $k = 1, \dots, K$ we measure the proportion of configurations that return to themselves after k steps,

$$m_k = \frac{|\{x \in C : F^k(x) = x\}|}{N}.$$

The length of the vector does not depend on the number of nodes but on the window K , so rules with different numbers of states, neighbors or spaces can be compared; K must be chosen so that it captures the relevant cycles of both graphs. Since m_k depends only on the eigenvalues, this method inherits the blindness to transients. With $K = 60$ we obtain a maximum of 74 classes. In the full graph, increasing K stabilizes the number of classes as n grows, while in the quotient K has little influence, since the cycles are shorter.

5.3 Spectral moments with attractor mass

To distinguish rules such as 0 and 8, we need to measure, besides the attractors, how strongly they attract the other configurations, that is, how many configurations reach each attractor and in how many steps. The idea is that each cycle is still activated every p steps, as in the spectral moments, but at each activation we also add the configurations that have already reached it. Using the height $h(x)$ and the period $p(x)$ of each configuration, we define

$$\tilde{m}_k = \frac{|\{x \in C : p(x) \mid k \text{ and } h(x) \leq k - p(x)\}|}{N}, \quad k = 1, \dots, K.$$

That is, a configuration counts at step k if its attractor is activated at that step and the configuration had already reached it before the previous activation: cycle nodes ($h = 0$) count from the first activation ($k = p$), nodes with $1 \leq h \leq p$ from the second ($k = 2p$), and so on. Since every periodic configuration satisfies the condition when $p \mid k$, the vector decomposes as $\tilde{m}_k = m_k + t_k$, where m_k is the spectral moment and t_k is the proportion of transient configurations that count at step k . Thus, the new vector extends the spectral moments with the information of the basins and keeps their length K independent of the domain, with the limitation that a cycle with $p > K$ is never activated.

For example, with 5 cells ($N = 32$), rule 0 sends every configuration to zero in one step ($h \leq 1$), whereas in rule 8 each block of ones leaves an isolated 1 that disappears in the next step ($h \leq 2$); of the 32 configurations, 12 reach zero in at most one step. Their vectors, $\tilde{m}^{(0)} = \frac{1}{32}(1, 32, 32, \dots)$ and $\tilde{m}^{(8)} = \frac{1}{32}(1, 12, 32, \dots)$, are no longer equal, even though their spectral moments are.

With the attractor mass the number of classes increases considerably (Figure 2): in the full graph, with $K = 60$ and $n \geq 6$ we obtain between 79 and 83 classes, and the quotient reaches up to 80. Rules 0 and 8 are separated for every $n \geq 3$, and combining $n = 3$ and $n = 6$ gives 86 classes with the full graph. Only two pairs remain together: $\{1, 19\}$, whose exact heights are different but fall in the same interval of p steps, and $\{4, 12\}$, which coincide in the distribution of height and period and differ only in the shape of the trees. The quotient, on the other hand, groups the rules that differ by a composition with the translation ($\{15, 51\}$, $\{170, 204\}$, $\{4, 12, 34\}$, $\dots$) and only reaches 81 classes. Using the full graph and the quotient on the same size gives 87 classes with $n = 6, 10, 12, 14, 18, 22, 24$ and 26.

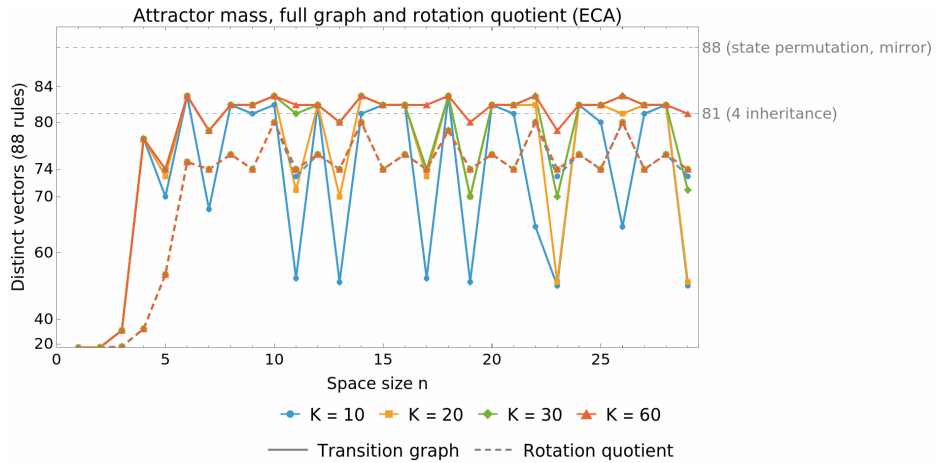


Fig. 2: Number of distinct spectral moment vectors with attractor mass among the 88 representative rules for $n = 1, \dots, 29$ and $K = 10, 20, 30, 60$, in the full graph (solid line) and in the quotient graph (dashed line).

Figure 1 and Table 1 compare all the methods. The canonical form distinguishes more classes, but the spectral moments with attractor mass are the second best method and can produce 87 classes in total, followed by the eigenvalues and the plain spectral moments.

Method	Full graph		Quotient		Full
	One size	All	One size	All	+ quotient
Canonical form	85	88	85	85	88
Eigenvalues	74	77	63	68	77
Spectral moments	74	77	63	68	77
Attractor mass	83	86	80	81	87

Table 1: Maximum number of classes among the 88 representative rules. “One size”: best n ; “All”: combines the 29 sizes; “Full + quotient”: both graphs.

6 Experiments

In this section we evaluate the spectral moments with attractor mass with $K = 60$ and the cosine distance $d(u, v) = 1 - u \cdot v / (\|u\| \|v\|)$, in the full graph and in the quotient. The vectors of the rules from other domains were computed with up to $n = 12$ cells and at most 3 states ($3^{12} = 531\,441$ configurations).

6.1 Related rules from different domains

We computed the four relations of Section 3 between all the rules of the domains $(q, m) = (2, 1), (2, 2), (2, 3), (3, 1)$ and $(3, 2)$, together with the 3-state extension of every ECA (domain $(3, 3)$), and we compare on the same space size the vectors of each pair of rules directly linked by a relation (Table 2).

With $m = 2$, the reflected neighborhood $(i - 1, i)$ is $(i, i + 1)$, so the mirror rule is the original one composed with a translation: the quotient graph ignores this effect, while the full graph is sensitive to the translation (median 0.54). The same happens with neighborhood reduction: if the remaining neighbors keep their position, the reduced rule has the same global function and the distance is 0, and if the neighborhood is shifted, the global function changes by a translation. The only ECA reductions with positive distance in the quotient are those of rules that depend on non-contiguous neighbors (σ_{i-1} and σ_{i+1}), because with even n the ring splits into two sublattices of $n/2$ cells; with odd n (for example, $n = 11$) the distance is 0.

State reduction never gives distance 0. The extension of a rule to $q + 1$ states preserves its cycles, but adds $(q + 1)^n - q^n$ Gardens of Eden, which with $n = 12$ make up 99.2% of the configurations. Even so, the extension largely preserves the order of the distances (Spearman correlation of 0.47 in the full graph and 0.76 in the quotient), although the original rule is the ECA closest to its extension in only 5 of the 88 cases (10 in

Relation	Domains	Pairs with $d = 0$			Median of d	
		Pairs	Full	Quotient	Full	Quotient
State permutation	all except (3, 3)	1010	1010	1010	0	0
Mirror, odd m	(2, 3), (3, 3)	384	384	384	0	0
Mirror, $m = 2$	(2, 2), (3, 2)	104	12	104	0.54	0
Neighborhood reduction	(2, 3) $\rightarrow$ (2, 2)	38	20	32	0	0
	(2, 2) $\rightarrow$ (2, 1)	6	4	6	0	0
	(3, 2) $\rightarrow$ (3, 1)	51	33	51	0	0
State reduction	(3, 1) $\rightarrow$ (2, 1)	4	0	0	0.008	0.008
	(3, 2) $\rightarrow$ (2, 2)	16	0	0	0.015	0.008
	(3, 3) $\rightarrow$ (2, 3)	256	0	0	0.083	0.010
ECA of different classes	(2, 3)	3820	6	11	0.33	0.10

Table 2: Distance between the attractor mass vectors of rules linked by an inheritance relation, with $n = 12$ and $K = 60$. The last row is the reference: pairs of ECA that do not belong to the same class.

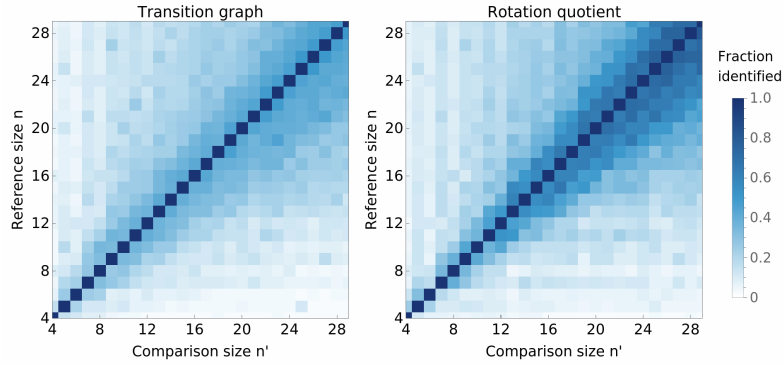


Fig. 3: Fraction of the 88 ECA that are identified between sizes n and n' ($K = 60$).

the quotient). In contrast, the plain spectral moments do give distance 0, because the extension has the same cycles and only the normalization changes.

Finally, the distance between rules 0 and 8 is positive for every $n \geq 3$ and grows with n (0.0072 with $n = 12$) until it stabilizes at 0.0085 with $n = 29$. The rules closest to 0 are not 8 but 4, 12 and 200 ($d \leq 0.0004$), which reach a fixed point in a single step.

6.2 One rule on spaces of different sizes

We compare each of the 88 ECA with itself over all pairs of sizes $n \neq n'$ with $n, n' \in \{8, \dots, 29\}$. The median distance between the vectors of the same rule is 0.098 in the full graph and 0.001 in the quotient, against 0.47 and 0.23 between different rules of the same size. However, since many rules have very close neighbors, a rule is identified (its nearest vector at the other size is its own) in only 25% of the cases in the full graph and 33% in the quotient (Figure 3).

Stability depends on the dynamics of each rule. Negation (51) and identity (204) are identified for every pair of sizes and, in the full graph, so are other rules whose vector barely changes with n , such as 0, 8, 54, 62 and 94, in more than 80% of the pairs. Shift rules (for example 2, 24, 34, 42, 56 and 170) are never identified in the full graph, because the length of their cycles depends on the divisors of n ; the quotient improves them (identification ranges from 25% of the pairs for rule 2 to all of them for rule 170), but the chaotic and additive rules (18, 60, 122, 126, 146 and 150) are almost never identified. Therefore, the comparison between different sizes is only reliable for rules whose attractors do not depend on n , and in general it is better to compare on the same size.

6.3 Clustering of the ECA

Finally, we cluster the 88 representative ECA with $n = 12$ and $K = 60$. We tried k -medoids and hierarchical clustering with 4 and 5 clusters; the best partition is the k -medoids one with 4 clusters (silhouette 0.70). Figure 4 shows a projection of the clusters onto two dimensions by classical multidimensional scaling, which preserves 77% of the variance. The clusters are separated by the dominant period of their attractors (Table 3): the first gathers rules that send almost everything to fixed points, with a vector that saturates close to 1, such as the null rules and the rules with many fixed points; the second, rules whose attractors shift, with cycles of period n or $2n$ that are only activated at $k = 12, 24, \dots$; the third, rules with period-2 attractors, whose vector alternates between even and odd steps, such as negation and the additive rules; and the fourth, rules with long cycles.

Cluster	Period	Rules
Fixed point (30)	1 (29)	0, 4, 8, 12, 13, 22, 30, 32, 36, 40, 44, 72, 76, 77, 78, 94, 104, 106, 108, 128, 132, 136, 140, 160, 164, 168, 172, 200, 204, 232
Period n (30)	12 (27)	2, 6, 9, 10, 11, 14, 15, 24, 25, 26, 34, 38, 41, 42, 43, 45, 46, 54, 56, 60, 74, 130, 134, 138, 142, 152, 154, 162, 170, 184
Period 2 (22)	2 (16)	1, 5, 7, 18, 19, 23, 28, 29, 33, 37, 50, 51, 57, 73, 90, 105, 122, 126, 146, 150, 156, 178
Long periods (6)	24 (3)	3, 27, 35, 58, 62, 110

Table 3: Clusters of Figure 4. Period: the dominant period, which gathers the most configurations within the window, and in parentheses the number of rules of the cluster in which it dominates.

However, the chaotic rules do not form a cluster of their own, for two reasons. In some of them, most configurations reach cycles with period greater than K , which are never activated, and almost all the others reach a fixed point; since the cosine distance discards the magnitude of the vector, they end up in the fixed-point cluster (e.g., rule 30). In others, most do reach a fixed point (e.g., rules 22 and 106).

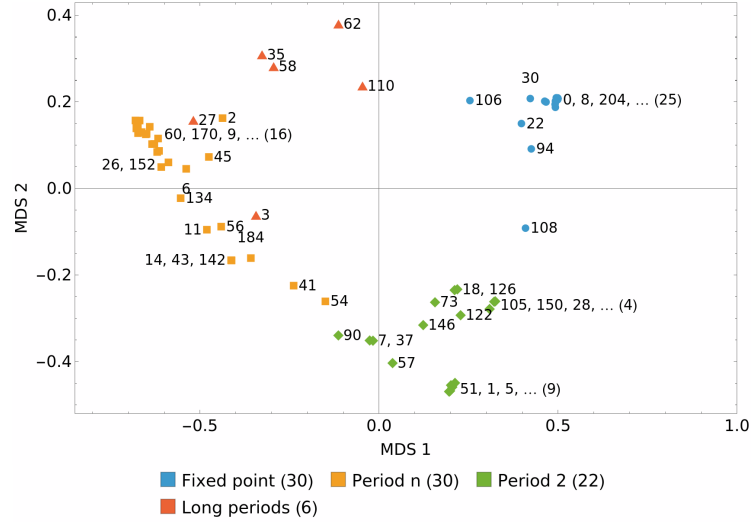


Fig. 4: Two-dimensional projection of the clustering of the 88 representative ECA by attractor mass ($n = 12$, $K = 60$, k -medoids with 4 clusters). Very close points share a label, with the number of rules in parentheses.

7 Conclusions

The four inheritance relations connect rules from different domains and reduce the 256 ECA to 81 classes. The canonical form of the transition graph exactly distinguishes the 88 classes under permutation and mirror with a single space of 6 cells if both the full graph and its quotient by translations are used, but it gives a binary comparison limited to rules of the same domain.

The eigenvalues and the spectral moments only see the attractors: they do not separate rules that differ in their transients, such as 0 and 8, and they stop at 77 classes. The spectral moments with attractor mass keep their fixed length and add the information of the basins: they distinguish 86 of the 88 classes (87 together with the quotient) and separate rules 0 and 8, although with a small distance (0.007 with $n = 12$) because the difference lies in the first steps of a vector that saturates.

Across domains, the attractor mass gives zero distance to the relations that preserve the transition graph and, in the quotient, also to those that change it by a translation; thus, making the neighbors contiguous seems to preserve the behavior (modulo translations, except for non-contiguous neighborhoods on spaces of even size), but the exact comparison is to keep the neighbors at their original positions: in that case the global function is the same and the distance is always zero, as when an end neighbor is removed. State reduction is not identified, because the configurations with the new state dominate the normalization; the plain spectral moments, in contrast, are invariant under it. Moreover, the vector depends on the size of the space, so it is better to compare on the same size. With that precaution, the distance groups the ECA into four families according to the

dominant period of their attractors, although the chaotic rules are scattered because their long cycles do not fit in the window.

Future work includes: normalizing the attractor mass so that it is invariant under state reduction, for example by restricting it to the configurations that survive the first step; giving more weight to the first steps and keeping the magnitude of the vector; choosing the space size and the window K systematically; checking whether the similarity holds when the neighbors are made contiguous; studying dimension reduction; and extending the comparison to larger domains through sampling.

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References

1. Torres, J.: Introducción a los sistemas complejos. Universidad de Granada (2025)
2. Solé, R.V., Manrubia, S.C.: Orden y caos en sistemas complejos. Edicions UPC, Barcelona (1996)
3. Hoekstra, A.G., Kroc, J., Slood, P.M.A.: Introduction to modeling of complex systems using cellular automata. In: Hoekstra, A.G., Kroc, J., Slood, P.M.A. (eds.) *Simulating Complex Systems by Cellular Automata*, pp. 1–16. Understanding Complex Systems, Springer, Berlin, Heidelberg (2010). https://doi.org/10.1007/978-3-642-12203-3_1
4. Wolfram, S.: Cellular automata as simple self-organizing systems. Technical Report CALT-68-938, California Institute of Technology (1982), <https://content.wolfram.com/sw-publications/2020/07/cellular-automata-self-organizing-systems.pdf>
5. Wolfram, S.: *A New Kind of Science*. Wolfram Media, Champaign, Illinois (2002)
6. Li, W., Packard, N.: The structure of the elementary cellular automata rule space. *Complex Systems* **4**(3), 281–297 (1990)
7. Martínez, G.J.: A note on elementary cellular automata classification. *Journal of Cellular Automata* **8**(3–4), 233–259 (2013)
8. LifeWiki: Conway’s Game of Life, [https://conwaylife.com/wiki/Conway’s_Game_of_Life](https://conwaylife.com/wiki/Conway's_Game_of_Life), last accessed 2026/08/28
9. Weisstein, E.W.: Cellular automaton. MathWorld – A Wolfram Resource (2002), <https://mathworld.wolfram.com/CellularAutomaton.html>, last accessed 2025/07/23
10. Sancho Caparrini, F.: Autómatas celulares. Universidad de Sevilla (2016), https://www.cs.us.es/~fsancho/Blog/posts/Automatas_Celulares.md.html, last accessed 2025/09/24
11. Bhattacharjee, K.: Cellular Automata: Reversibility, Semi-Reversibility and Randomness. Ph.D. thesis, Indian Institute of Engineering Science and Technology, Shibpur (2019), <https://arxiv.org/abs/1911.03609>, last accessed 2025/08/23
12. Ureña Ponce, O.: Diseño de un procesador de autómatas celulares para la simulación de sistemas complejos. Master’s thesis, Universidad de Guadalajara, Guadalajara, México (2015), <https://riudg.udg.mx/bitstream/20.500.12104/79915/1/MCUCEI01068FT.pdf>